

DATA-260 - Homework 1 Report

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Section 0 - Configuration

Value	Result
SID4	3170
PORT_BASE	8470 (8000 + 3170 mod 900)
PREFIX	s3170
SEED	3170
VERIFY_SEED	263170 (260000 + 3170)
DOMAIN_ID	2 -> Municipal Transit Incidents (3170 mod 8)
Hardware	MacbookPro15,1 Intel Core i9 16GB
Local model used	qwen3:8b (reasoning disabled) via Ollama
Tagged commit hash	hw1
GitHub repo link	https://github.com/pragnayap/data260-3170

Step 1 - Domain Assignment & Schema

DOMAIN_ID = 2 -> Municipal Transit Incidents. Entity fields and category values were defined in DOMAIN_SCHEMA.md before writing any HTML, per the assignment's 'before coding' requirement.

Entity fields (from DOMAIN_SCHEMA.md)

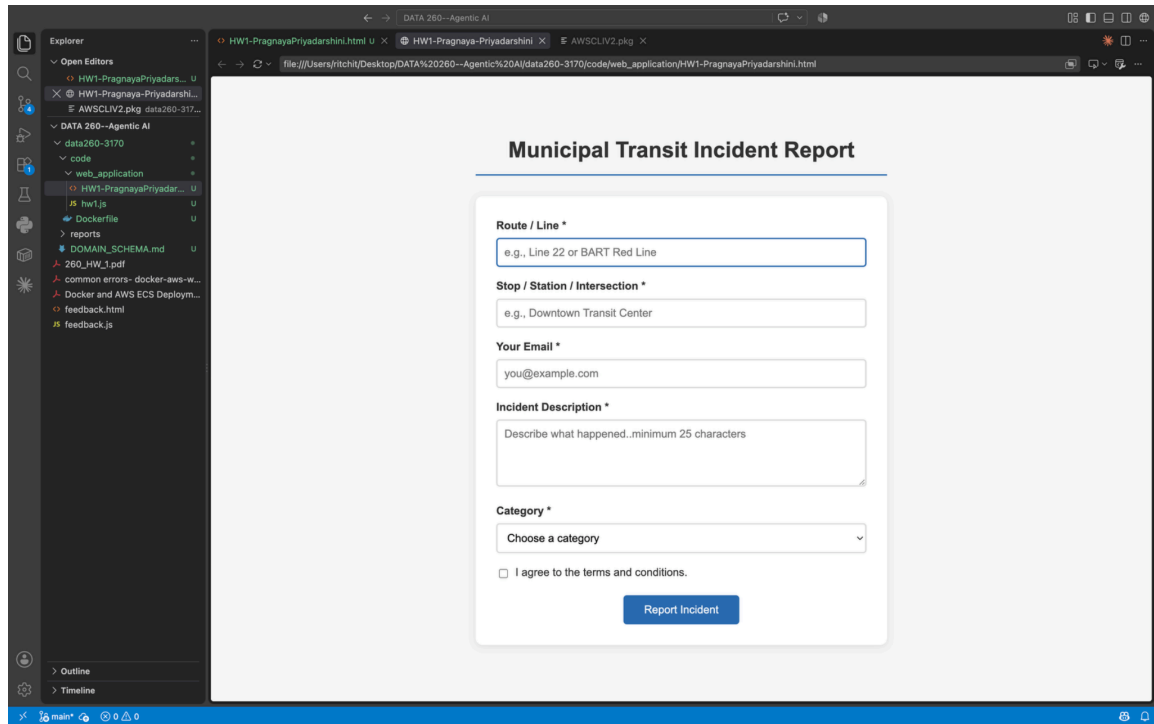
Field	Type	Notes
routeId (primary)	text, required, autofocus	e.g. 'Line 22' or 'BART Red Line'
location (secondary)	text, required	Stop/station/intersection
submitterEmail	email, required	Reporter's email
description (content)	textarea, required	> 25 characters
category	select, 1 of 4	Delay / Mechanical Failure / Accident-Collision / Safety Hazard
agreeTerms	checkbox, required	'I agree to the terms and conditions.'

Step 2 - HTML Form (Part I)

File: code/web_application/HW1-PragnayaPriyadarshini.html

Build the Municipal Transit Incident Report form with all required fields, the primary field set to autofocus on page load, and a 4-option category dropdown.

Screenshot: fresh page load (autofocus on Route/Line field)

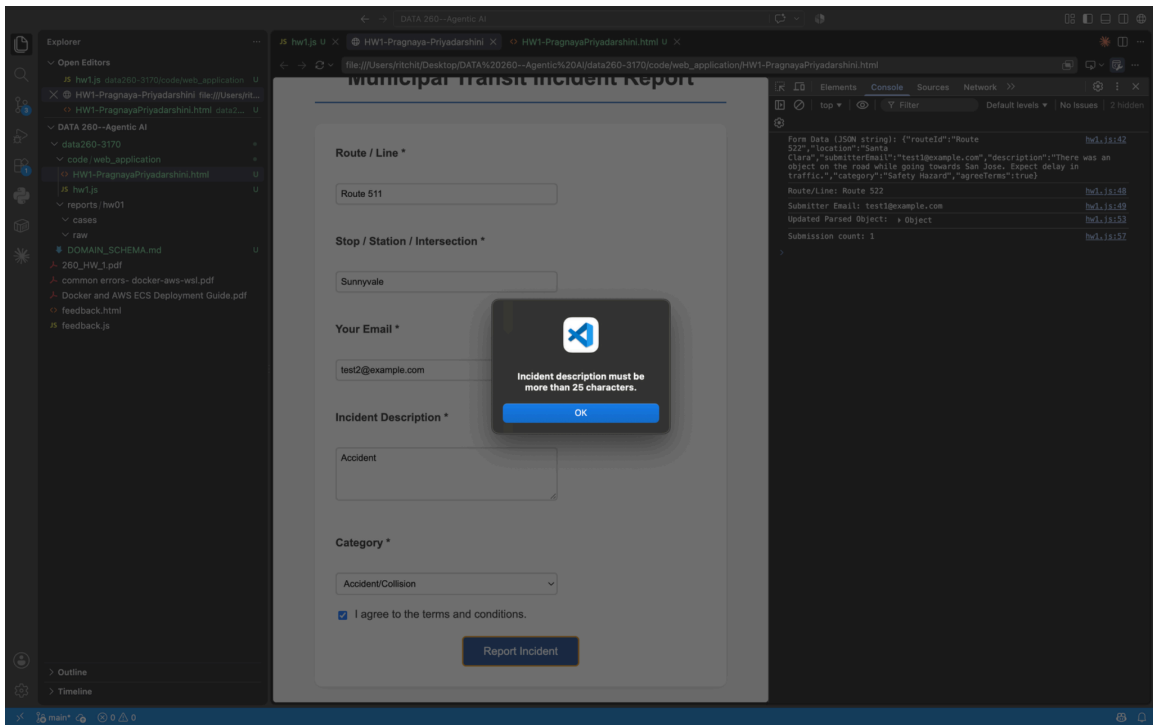


Step 3 - JavaScript Validation (Part II)

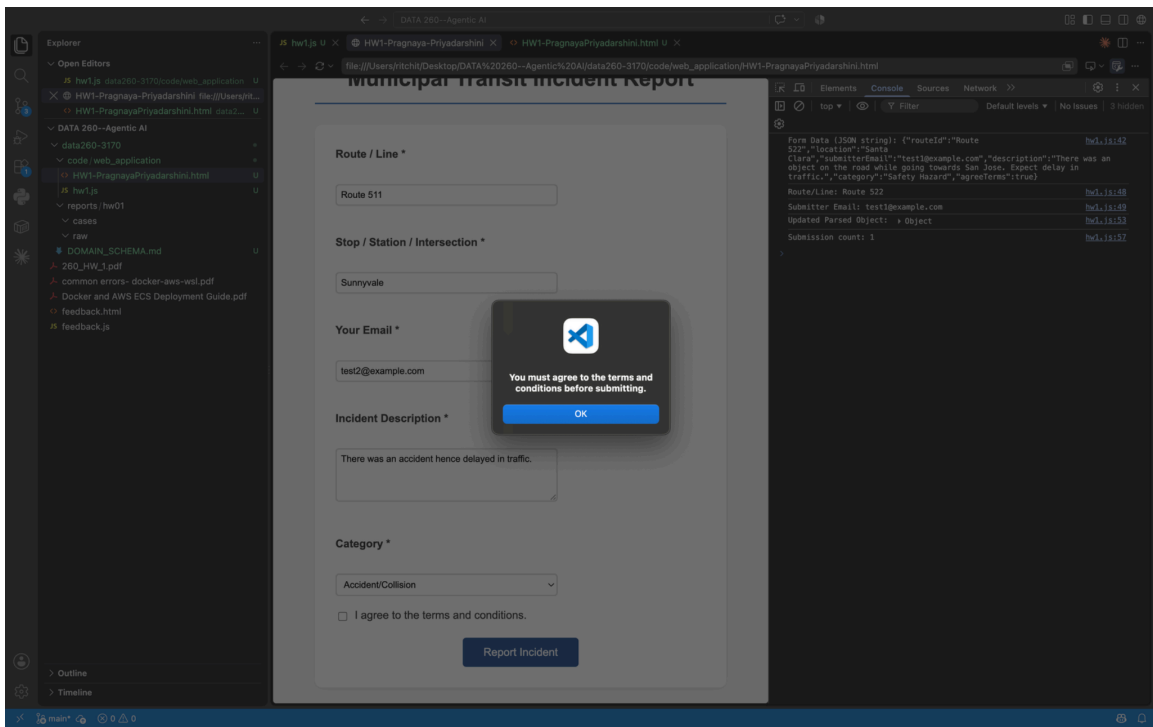
File: code/web_application/hw1.js

Implemented: arrow-function validation (description length, checkbox), JSON.stringify/parse of form data, destructuring of the primary + email fields, spread operator to add submissionDate, and a closure-based submission counter.

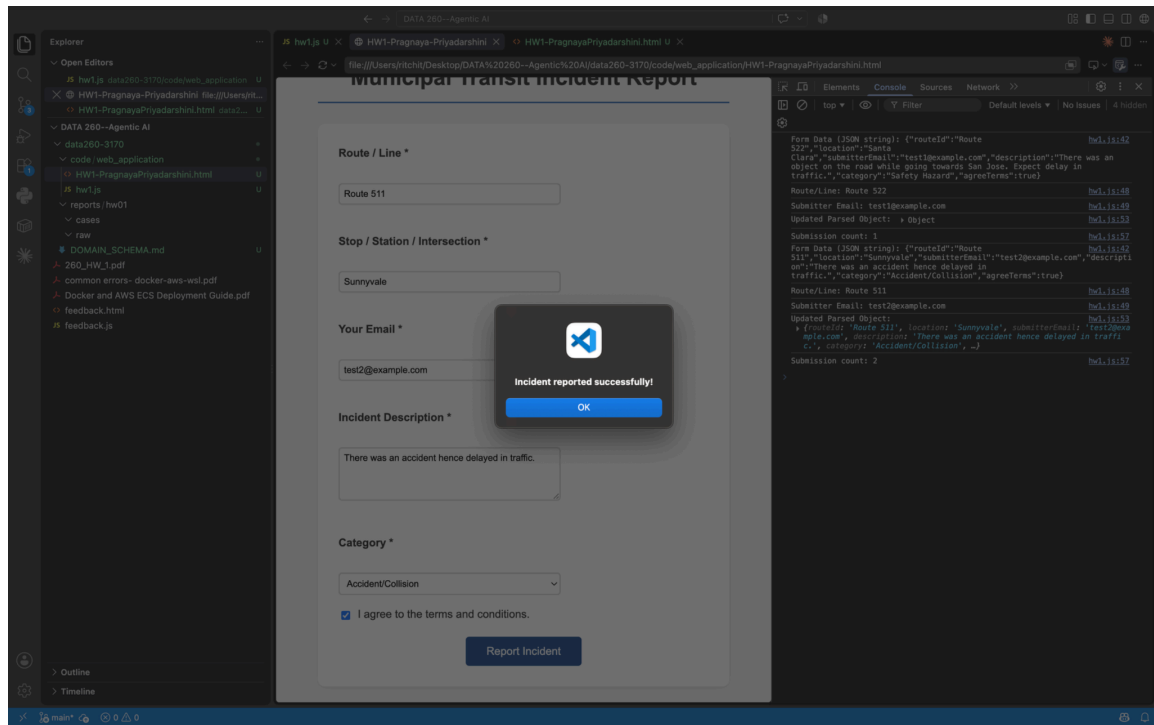
Screenshot: validation alert - description too short (<=25 characters)



Screenshot: validation alert - terms checkbox unchecked



Screenshot: console output after a successful submission



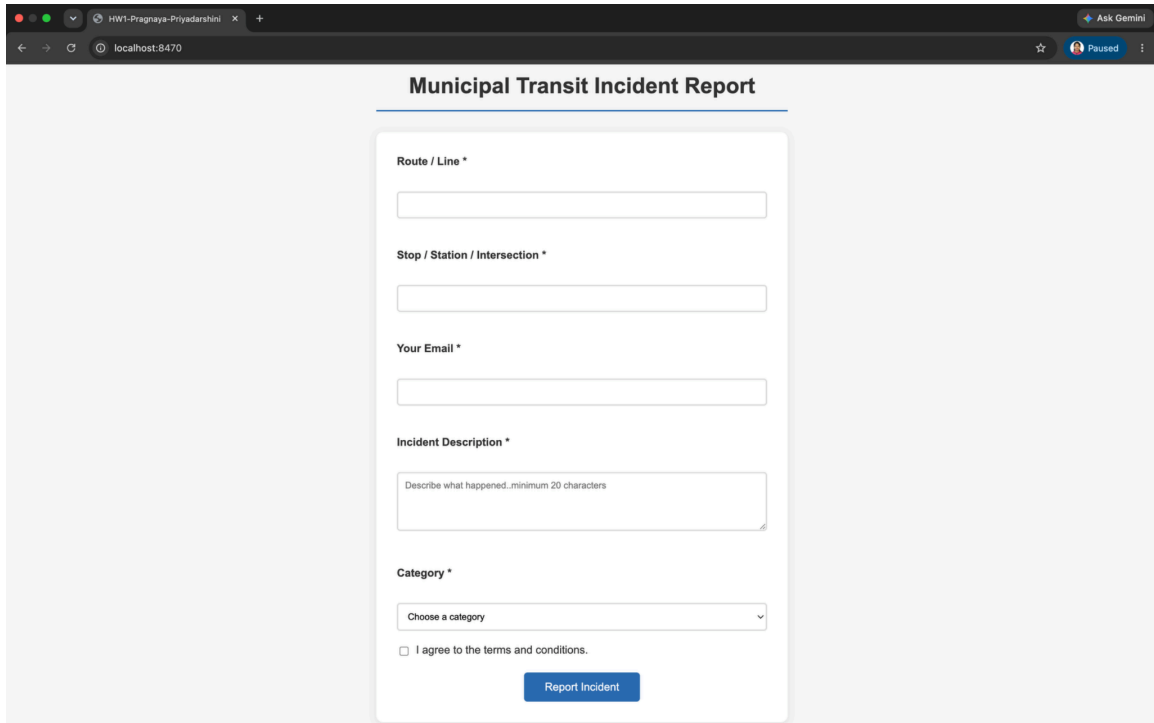
Step 4 - Docker (local deployment)

File: code/Dockerfile - nginx:alpine base image, reconfigured via sed to listen on PORT_BASE (8470) and serve HW1-PragnayaPriyadarshini.html as the index file.

```
docker build -t s3170-hw1-web
```

```
docker run -d -p 8470:8470 --name s3170-hw1-web-run s3170-hw1-web
```

Screenshot: App running on localhost:8470



Municipal Transit Incident Report

Route / Line *

Stop / Station / Intersection *

Your Email *

Incident Description *

Describe what happened .minimum 20 characters

Category *

Choose a category

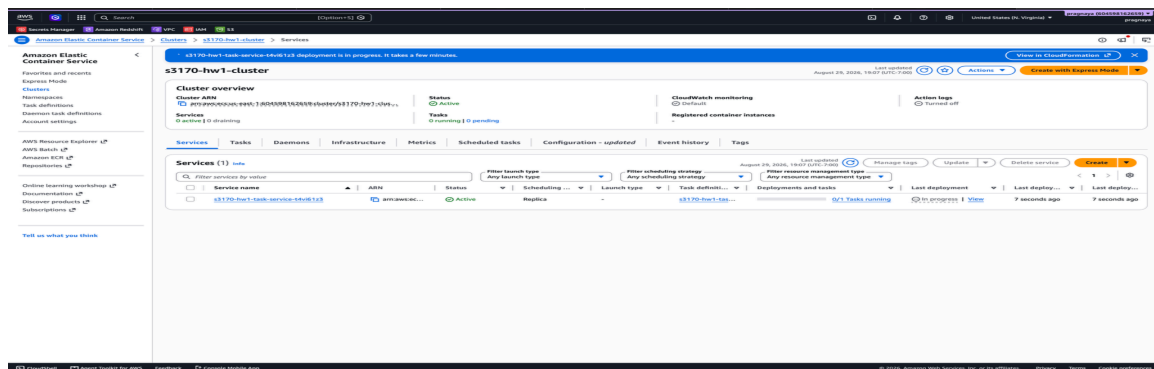
☐ I agree to the terms and conditions.

Report Incident

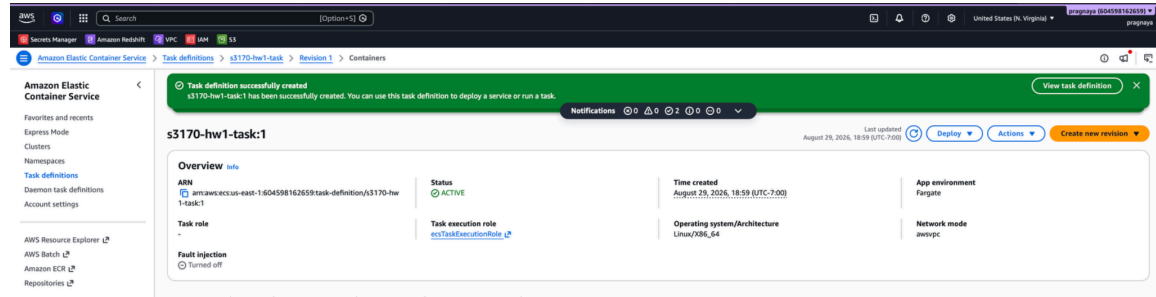
Step 5 - AWS ECS Deployment

The web image(s3170-hw1-web) was pushed to ECR, creating a fargate cluster as attached screenshot below (s3170-hw1-cluster) and a 1-task service (s3170-hw1-service).

Cluster:



Service:

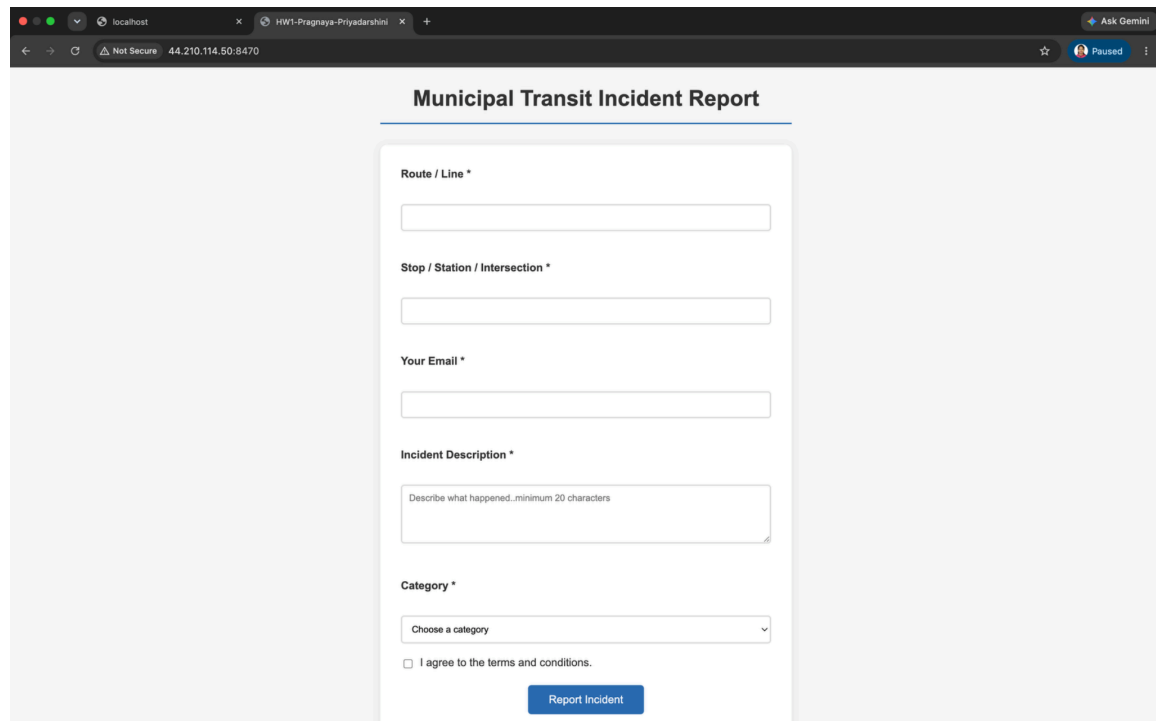


Issue encountered and fixed

The security group (s3170-hw1-sg) initially had no inbound rule, and the outbound rule was restricted to only port 8470 - which was blocking the fargate task from reaching ECR's API over HTTPS (443) to authenticate and pull the image.

Which was fixed by adding an inbound rule for TCP 8470 from 0.0.0.0/0 and replacing the restrictive outbound rule with an allow all rule. And then was verified the fix by doing a new deployment and confirming the task has reached RUNNING status.

Screenshot: app running on public IP:8470



Step 6 - Agentic AI (Part 2)

File: code/agents_demo.py - Planner -> Reviewer -> Finalizer pipeline using ChatOllama (qwen3:8b, reasoning disabled, strict JSON output). Tags/summary are derived only from the input title/content (no hardcoded domain keywords).

Command used

```
python agents_demo.py --title "Bus number 522 breaks down in Sunnyvale, near MollyTea." \
```

```
--content "A bus bearing bus number 522 which was going from Palo alto in El Camino Real towards San Jose, suddenly faced a technical failure..." --temperature 0.0
```

Screenshots: Planner / Reviewer / Finalized output

```
data60-3178/code on } main [?] via v3.12.13 (.venv) on (us-east-1) took 3m3s
python agents_demo.py --title "Bus number 522 breaks down in Sunnyvale, near MollyTea." --content "A bus bearing bus number 522 which was going from Palo alto in El Camino Real towards San Jose, suddenly faced a technical failure for which the lane got blocked and traffic didnot move. It was on the road blocking other vehicles till the Fire & Towing vehicle came. Which was really frustrating to everyone on road. After the emergency vehicle cleared the road after nearly 2 hours the traffic got cleared around 6p.m in the evening." --temperature 0.0

--- Planner (41227 ms) ---
{
  "thought": "The incident involves a bus breakdown causing traffic disruption and requiring emergency response.",
  "message": "Bus breakdown on El Camino Real caused traffic delays and required fire and towing assistance.",
  "data": {
    "tags": [
      "bus breakdown on el camino real",
      "traffic disruption in sunnyvale",
      "emergency vehicle response"
    ],
    "summary": "A bus breakdown on El Camino Real caused significant traffic delays in Sunnyvale.",
    "issues": [
      "traffic congestion",
      "public transportation disruption",
      "emergency response time"
    ]
  }
}

--- Reviewer (26761 ms) ---
{
  "thought": "The incident involved a bus breakdown on El Camino Real, causing traffic delays and requiring emergency assistance.",
  "message": "A bus breakdown on El Camino Real caused traffic delays and required fire and towing assistance.",
  "data": {
    "tags": [
      "bus breakdown",
      "traffic delay",
      "emergency response"
    ],
    "summary": "A bus breakdown on El Camino Real caused traffic delays and required emergency assistance.",
    "issues": []
  }
}

Finalized Output
{
  "thought": "Finalizing tags and summary based on reviewer feedback and input details.",
  "message": "Bus breakdown on El Camino Real caused traffic delays and required fire and towing assistance.",
  "data": {
    "tags": [
      "bus breakdown",
      "traffic delay",
      "emergency response"
    ],
    "summary": "A bus breakdown on El Camino Real caused significant traffic delays and required emergency assistance.",
    "issues": []
  }
}

Publish Package
{
  "title": "Bus number 522 breaks down in Sunnyvale, near MollyTea.",
  "email": "student@example.com",
  "content": "A bus bearing bus number 522 which was going from Palo alto in El Camino Real towards San Jose, suddenly faced a technical failure for which the lane got blocked and traffic didnot move. It was on the road blocking other vehicles till the Fire & Towing vehicle came. Which was really frustrating to everyone on road. After the emergency vehicle cleared the road after nearly 2 hours the traffic got cleared around 6p.m in the evening.",
  "agents": {
    "transcript": [
      {
        "role": "Planner",
        "content": "Bus breakdown on El Camino Real caused traffic delays and required fire and towing assistance."
      },
      {
        "role": "Reviewer",
        "content": "A bus breakdown on El Camino Real caused traffic delays and required fire and towing assistance."
      }
    ]
  },
  "final": {
    "tags": [
      "bus breakdown",
      "traffic delay",
      "emergency response"
    ],
    "summary": "A bus breakdown on El Camino Real caused significant traffic delays and required emergency assistance.",
    "issues": []
  },
  "submissionDate": "2025-08-30T07:48:26Z"
}
}

(data60-3178/code on } main [?] via v3.12.13 (.venv) on (us-east-1) took 3m3s
```

Short answers to the input questions in the terminal:

Q1. The 3 final tags obtained:

Bus breakdown, Traffic delay, emergency response

Q2. The final summary (<=25 words):

A bus broke down on el camino real causing significant traffic delays and requiring emergency assistance.

Q3. Did the Reviewer agent change anything?

Yes - the Planner proposed verbose, descriptive tags ("bus breakdown on el camino real", "traffic disruption in sunnyvale", "emergency vehicle response") and listed 3 issues; the Reviewer simplified these to short, consistent tags ("Bus breakdown", "Traffic delay", "emergency response") and cleared the issues list to empty, keeping the same underlying topics but tightening the phrasing.

Q4. Explanation how the Planner, Reviewer and Finalizer worked together:

- The planner looks at the input as "the bus broke in the El Camino which requires emergency service and there will be the traffic delay at Sunnyvale" and it proposes tags and issues.
- The reviewer takes the planner's output and separates the useful tags from filler and simplifies the tags and clears the issues list.
- The finalizer then finalises the reviewer's output based on tags and gives the final result.

Step 7 - Measuring Non-Determinism (Part 3)

Fixed input was saved once at reports/hw01/cases/nondeterminism_input.json. Then ran the python "code/run_nondeterminism.py" file. First 20 runs at temperature 0.7, then 20 runs at temperature 0.0 (40 total), using the same Planner—>Reviewer—>Finalizer pipeline as Part 2.

```
caffeinate -i python run_nondeterminism.py 2>&1 | tee -a ../reports/hw01/RUN_LOG.txt
```

Tag Stability

Metric	Temp 0.7	Temp 0.0
Distinct tag sets	5	1
Tags in all 20 runs	none	emergency response, public transportation, traffic incident
Tags in exactly 1 run	emergency assistance, public transportation, traffic incident	none

Latency (ms)

Metric	Temp 0.7	Temp 0.0
p50	76,921.5	72,597.0
p95	82,781.4	73,167.2
p99	82,985.9	80,603.8

Observations are explained below,

1. At temp 0.0, the data showed 1 distinct tag set throughout all the 20 runs. So 2 users who submit identical incidents would see exactly the same 3 tags and summary everytime. Hence our data showed 1 distinct tag across all 20 runs with all 100% tags appearing in all runs. This happens because the model always chooses no randomness and picks the most probable single word. It goes with whatever is the highest scored word. Then if nothing changes in the input it keeps the same exact choices and ends up giving the same output. That's why all 20 temp 0.0 runs are having 3 identical tags.
2. At temp 0.7, 2 users who submit identical data would likely see different tag sets. The result showed 5 tag sets across all 20 runs with 3 tags appearing in only a single run each. While the core theme stayed similar. While the correctness did not change but while browsing the tags would fall under transit incidents's reasonable category.
3. We can accept the variance if the reported transit incident is slightly different and the tags across runs does not change the correctness of the system. Because the user browsing by tags would still find the incident under a reasonable category either way.
4. We can not accept the variance when the reported incidents are different but the identical inputs producing different outputs in two different runs would be wrong and can lead to a serious safety problem.

Step 8 - Model Client and Token Accounting (Part 4)

File: `src/model_client.py` - reusable ModelClient adapter with a complete(messages, tools=None) interface; centralizes per-turn and cumulative token accounting. File: `code/hw1_client.py` - CLI demo that imports the adapter, loads AGENT.md as the system prompt, and supports `/stats` and `/exit`.

AGENT.md verification

AGENT.md asked the model to respond to code reviews using ONLY a bulleted list. Verified by asking it to review a snippet with a real bug (adding a number and a string); it replied with exactly one bullet, correctly identifying the type coercion issue, with no prose outside the bullet.

Screenshot: 5-turn conversation with /stats after turn 3 and turn 5

```
data268-3178/code on [?] via v3.12.13 (.venv) on (us-east-1) took 4ms
[?] python hw1_client.py
Type your message, '/stats' for stats, or '/exit' to quit.

You: What is a bus route?
[turn 1] input_tokens=116 output_tokens=22 total_tokens=139
Assistant: - A bus route is a predefined path or line that a bus follows to transport passengers between stops or stations.

You: What is a transit incident?
[turn 2] input_tokens=154 output_tokens=34 total_tokens=188
Assistant: - A transit incident is an unexpected event that disrupts normal operations of a public transportation system, such as delays, service suspensions, accidents, or equipment failures.

You: Review this code: function add(a, b) { return a + b } const result = add(5, "3");
[turn 3] input_tokens=223 output_tokens=38 total_tokens=253
Assistant: - The code attempts to add a number and a string, which will result in a string "53" due to JavaScript's type coercion.

You: You: /stats
[stats] {'turn_count': 3, 'cumulative_input_tokens': 493, 'cumulative_output_tokens': 87, 'cumulative_total_tokens': 580, 'history_length_chars': 1215}
You: What typically causes a Delay versus a Safety Hazard in transit incidents?
[turn 4] input_tokens=275 output_tokens=54 total_tokens=329
Assistant: - A **Delay** is typically caused by operational issues like mechanical failures, traffic congestion, or scheduling problems.
- A **Safety Hazard** is typically caused by risks to passenger or staff safety, such as equipment malfunctions, track failures, or security threats.

You: Summarize why maintenance towing might take longer during rush hour.
[turn 5] input_tokens=351 output_tokens=48 total_tokens=399
Assistant: - Maintenance towing may take longer during rush hour due to increased traffic congestion, reduced road space, and higher demand for road usage, which can slow down the movement of tow trucks and emergency vehicles.

You: /stats
[stats] {'turn_count': 5, 'cumulative_input_tokens': 1119, 'cumulative_output_tokens': 181, 'cumulative_total_tokens': 1300, 'history_length_chars': 1992}
You: /exit
[exit] turns=5 cumulative_input_tokens=1119 cumulative_output_tokens=181 cumulative_total_tokens=1300
(base)
data268-3178/code on [?] via v3.12.13 (.venv) on (us-east-1) took 2ms
```

Written answers (Kindly refer README.md for full text)

Why is prior conversation context resent with every turn?

LLM APIs are stateless between requests, so the full history must be resent for the model to respond consistently with earlier turns.

How is a system prompt different from a user message?

A system prompt sets standing instructions for the whole conversation; a user message is one turn of content to respond to. AGENT.md's bullet-only rule applied to every later turn, confirming this.

Why do input tokens grow over a conversation?

Each turn appends the previous user message and the model's reply into the resent history. Observed directly: input tokens climbed 116 -> 154 -> 223 -> 275 -> 351 across the 5 turns.

What eventually limits that growth?

The model's context window (num_ctx=2048 tokens).

Step 9 - Verification

code/verify_hw01.py runs 16 automated checks (required files exist, Python modules compile, the non-determinism raw data has all 40 records, Ollama is reachable and qwen3:8b is pulled). All 16 checks passed. Output saved to reports/hw01/verification.json.

AI Use

The full disclosure is in reports/hw01/AI_USE.md: what AI was used for vs. done independently, the qwen3:8b thinking-mode discovery (verified first-hand by timing the same prompt with and without reasoning disabled), how it was detected, and what changed as a result.